

## SAFETY DATA SHEET

Creation Date 24-June-2009

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** Aluminium-nickel, Raney-type alloy, powder

**Cat No. :** AC206230000; AC206230025; AC206230050; AC206231000; AC206235000

**CAS-No** 12635-29-9  
**Synonyms** Raney-type alloy for the preparation of Raney Nickel catalyst.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                        |             |                    |
|------------------------------------------------------------------------|-------------|--------------------|
| Substances/mixtures which, in contact with water, emit flammable gases | Category 2  | Gas(es) = Hydrogen |
| Respiratory Sensitization                                              | Category 1  |                    |
| Skin Sensitization                                                     | Category 1  |                    |
| Carcinogenicity                                                        | Category 1A |                    |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

In contact with water releases flammable gas  
May cause an allergic skin reaction  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
May cause cancer



### Precautionary Statements

#### Prevention

Wear protective gloves/protective clothing/eye protection/face protection  
Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Do not allow contact with water  
Handle under inert gas. Protect from moisture  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear respiratory protection

#### Response

IF exposed or concerned: Get medical advice/attention  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Brush off loose particles from skin. Immerse in cool water/wrap in wet bandages  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
If experiencing respiratory symptoms: Call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

#### Storage

Store locked up  
Store in a dry place. Store in a closed container

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component                 | CAS-No     | Weight % |
|---------------------------|------------|----------|
| Nickel alloy, base, Ni,Al | 12635-29-9 | <=100    |

## 4. First-aid measures

|                       |                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b> | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>     | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>      | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |

|                                        |                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                              |

## 5. Fire-fighting measures

|                                         |                               |
|-----------------------------------------|-------------------------------|
| <b>Suitable Extinguishing Media</b>     | Dry sand.                     |
| <b>Unsuitable Extinguishing Media</b>   | DO NOT USE WATER, FOAM OR CO2 |
| <b>Flash Point</b>                      | No information available      |
| <b>Method -</b>                         | No information available      |
| <b>Autoignition Temperature</b>         | Not applicable                |
| <b>Explosion Limits</b>                 |                               |
| <b>Upper</b>                            | No data available             |
| <b>Lower</b>                            | No data available             |
| <b>Sensitivity to Mechanical Impact</b> | No information available      |
| <b>Sensitivity to Static Discharge</b>  | No information available      |

### Specific Hazards Arising from the Chemical

Water reactive. Contact with water liberates extremely flammable gases. Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors. Nickel oxides. Fumes of aluminum or aluminum oxide. Hydrogen.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
2

**Instability**  
2

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                                                  |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.                                                                |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                     |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.                              |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from water or moist air. Keep under nitrogen. Flammables area. Incompatible Materials. Strong oxidizing agents. Strong acids. |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                 | Alberta                    | British Columbia | Ontario TWAEV              | Quebec                     | ACGIH TLV | OSHA PEL                           | NIOSH IDLH                                                 |
|---------------------------|----------------------------|------------------|----------------------------|----------------------------|-----------|------------------------------------|------------------------------------------------------------|
| Nickel alloy, base, Ni,Al | TWA: 0.2 mg/m <sup>3</sup> |                  | TWA: 0.2 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> |           | (Vacated) TWA: 1 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.015 mg/m <sup>3</sup> |

**Legend**

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                     |                          |
|---------------------|--------------------------|
| Physical State      | Powder Solid             |
| Appearance          | Grey                     |
| Odor                | Odorless                 |
| Odor Threshold      | No information available |
| pH                  | No information available |
| Melting Point/Range | 1350 °C / 2462 °F        |
| Boiling Point/Range | No information available |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 3.460                    |
| Solubility                             | Insoluble in water       |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | Not applicable           |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Al Ni                    |
| Molecular Weight                       | 85.68                    |

## 10. Stability and reactivity

|                                  |                                                                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                    |
| Stability                        | Water reactive. Moisture sensitive.                                                                                                    |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Excess heat. Exposure to moist air or water.                                              |
| Incompatible Materials           | Strong oxidizing agents, Strong acids                                                                                                  |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors, Nickel oxides, Fumes of aluminum or aluminum oxide, Hydrogen |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                               |
| Hazardous Reactions              | None under normal processing.                                                                                                          |

## 11. Toxicological information

### Acute Toxicity

|                                                                                                   |                                                               |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Product Information                                                                               | The toxicological properties have not been fully investigated |
| Component Information                                                                             |                                                               |
| Toxicologically Synergistic Products                                                              | No information available                                      |
| <u>Delayed and immediate effects as well as chronic effects from short and long-term exposure</u> |                                                               |

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| Irritation      | May cause skin, eye, and respiratory tract irritation  |
| Sensitization   | May cause sensitization by inhalation and skin contact |
| Carcinogenicity | Limited evidence of a carcinogenic effect.             |

| Component                 | CAS-No     | IARC       | NTP   | ACGIH      | OSHA       | Mexico     |
|---------------------------|------------|------------|-------|------------|------------|------------|
| Nickel alloy, base, Ni,Al | 12635-29-9 | Not listed | Known | Not listed | Not listed | Not listed |

NTP: (National Toxicity Program)

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

|                      |                           |
|----------------------|---------------------------|
| Mutagenic Effects    | No information available  |
| Reproductive Effects | No information available. |

|                                                   |                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                             |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                             |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                            |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                            |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                              |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                              |
| <b>Other Adverse Effects</b>                      | No information available.                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Do not flush into surface water or sanitary sewer system. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Insoluble in water May persist                                        |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Is not likely mobile in the environment due its low water solubility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN3208                       |
| <b>Proper Shipping Name</b> | WATER-REACTIVE SOLID, N.O.S. |
| <b>Technical Name</b>       | Nickel alloy, base , Ni,Al   |
| <b>Hazard Class</b>         | 4.3                          |
| <b>Packing Group</b>        | II                           |

### TDG

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN3208                       |
| <b>Proper Shipping Name</b> | WATER-REACTIVE SOLID, N.O.S. |
| <b>Hazard Class</b>         | 4.3                          |
| <b>Packing Group</b>        | II                           |

### IATA

|                             |                                           |
|-----------------------------|-------------------------------------------|
| <b>UN-No</b>                | UN3208                                    |
| <b>Proper Shipping Name</b> | Metallic substance, water-reactive, n.o.s |
| <b>Hazard Class</b>         | 4.3                                       |
| <b>Packing Group</b>        | II                                        |

### IMDG/IMO

|                             |                                           |
|-----------------------------|-------------------------------------------|
| <b>UN-No</b>                | UN3208                                    |
| <b>Proper Shipping Name</b> | Metallic substance, water-reactive, n.o.s |
| <b>Hazard Class</b>         | 4.3                                       |
| <b>Packing Group</b>        | II                                        |

## 15. Regulatory information

### International Inventories

| Component                 | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|---------------------------|------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| Nickel alloy, base, Ni,Al | 12635-29-9 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                 | CAS-No     | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------|------------|-------|------|------|------|------|------|-------|-------|
| Nickel alloy, base, Ni,Al | 12635-29-9 | -     | -    | -    | -    | X    | -    | X     | -     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                 | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|---------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Nickel alloy, base, Ni,Al | Part 1, Group A Substance                            |                                                                            |                                           |

**Legend**

NPRI - National Pollutant Release Inventory

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component                 | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nickel alloy, base, Ni,Al | -                                                                   | Use restricted. See item 27. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                 | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Nickel alloy, base, Ni,Al | 12635-29-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                 | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Nickel alloy, base, Ni,Al | 12635-29-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

|                         |                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Creation Date</b>    | 24-June-2009                                                                                                                                                                       |
| <b>Revision Date</b>    | 24-December-2021                                                                                                                                                                   |
| <b>Print Date</b>       | 24-December-2021                                                                                                                                                                   |
| <b>Revision Summary</b> | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**